2026 Connecticut Sports Analytics Symposium Data Challenge

Optimal Timing of the Power Play in Mixed Doubles Curling

Abstract

In mixed doubles curling, the Power Play can be used once per game when a team has hammer, making its timing a key strategic choice. Using end-level data from 344 matches, we model Power Play timing as a sequential decision problem with stochastic scoring outcomes. A dynamic programming framework combined with Bayesian estimation is used to compute win probabilities and optimal Power Play policies across game states. We find that teams systematically delay Power Play usage relative to model recommendations, especially when trailing early, despite many recommendations being robust to data uncertainty.

1 Introduction and Strategic Context

In mixed doubles curling, each team is permitted to activate the *Power Play* exactly once per game, provided it has the hammer in that end. Activating the Power Play relocates the two pre-positioned stones away from the center line, substantially altering the geometry of the scoring area and the types of outcomes that become likely. While the rule governing Power Play usage is simple, deciding *when* to use it is one of the most consequential strategic choices in a match.

At an intuitive level, the Power Play creates a classic timing dilemma. Using it early may create opportunities to build a lead, but doing so sacrifices the option of deploying it later, when a single high-scoring end could decisively change the outcome of the game. Conversely, saving the Power Play for late ends may increase its leverage, but only if the game remains close and the team still has hammer when the opportunity arises. The value of the Power Play therefore depends on multiple interacting factors, including the current score differential, the number of ends remaining, hammer possession, and whether either team has already used its Power Play.

This interaction makes Power Play timing a *sequential decision problem under uncertainty*. Teams must choose whether to deploy the Power Play without knowing exactly how the current end will unfold, and each choice influences not only the immediate score but also the strategic landscape of future ends. A decision that is optimal in one game situation may be suboptimal or even detrimental in another.

Prior quantitative studies of curling have demonstrated that match outcomes can be modeled probabilistically using state variables such as score differential, hammer possession, and remaining ends. Early work by Willoughby and Kostuk showed that specific curling decisions can be analyzed using decision-tree and Markov-style reasoning [6, 7]. More recent studies have estimated win probabilities as functions of game state using Markov models [4, 2]. These approaches provide valuable tools for evaluating game situations, but they are primarily descriptive: they assess how likely a team is to win given a state, rather than prescribing which strategic action should be taken.

The goal of this project is to provide a *prescriptive, data-driven framework* for Power Play timing in mixed doubles curling. We treat a curling match as a sequence of ends, each producing an uncertain scoring outcome that moves the game into a new state. At each end in which a team has hammer and its Power Play available, the team faces a strategic choice between normal play and activating the Power Play. The quality of this choice is evaluated in terms of its long-run impact on the probability of winning the match.

Conceptually, our approach evaluates decisions by working backward from the end of the game. For every possible game situation, defined by the current score, hammer possession, remaining ends, and Power Play availability, we compute the probability of ultimately winning the match assuming optimal play by both teams. This backward-looking evaluation allows us to compare the long-term consequences of using the Power Play immediately versus saving it for later. The formal mathematical formulation of this decision process draws on classical dynamic programming and stochastic game ideas and is provided in Appendix A [1, 5].

A second key feature of our approach is the explicit treatment of statistical uncertainty. End-level scoring outcomes under normal play and under the Power Play are estimated from historical data and are therefore subject to sampling variability. Rather than relying on single point estimates, we adopt a Bayesian framework that treats these outcome distributions as uncertain and propagates this uncertainty through the decision model [3]. This allows us to assess not only which actions are optimal, but also how confident teams can be in those recommendations. Technical details of the

Bayesian modeling and uncertainty propagation are provided in Appendix B.

Taken together, this framework enables a principled comparison between historically observed Power Play usage and model-recommended decisions across the full space of game situations. Rather than presenting exhaustive policy surfaces in the introduction, we summarize here a small number of high-level strategic takeaways that recur throughout the analysis and are revisited in greater detail in subsequent sections.

The analysis reveals several consistent patterns in optimal Power Play timing:

- **Earlier usage when trailing:** The model frequently recommends activating the Power Play earlier than teams typically do in observed games, particularly when trailing by one to three points with multiple ends remaining (e.g., $t = 4$, $s = -2$ or $t = 5$, $s = -3$), where historical usage is rare despite substantial potential gains.
- **Asymmetry when ahead versus behind:** When trailing, earlier Power Play activation is often optimal, whereas when leading late in the game (e.g., $t = 2$, $s = 3$), the model generally favors restraint, in contrast to observed behavior.
- **Central role of hammer:** The largest discrepancies between recommended and observed decisions occur in hammer-team states, highlighting that the Power Play is most valuable as a tool for exploiting hammer advantage rather than as a last-resort option.
- **Robust versus ambiguous decisions:** Many recommended actions occur in states where the posterior probability of optimality exceeds 0.9, indicating stable guidance, while other states exhibit genuine strategic ambiguity and warrant greater flexibility.

2 Data and Empirical Setup

Our analysis uses the play-by-play data provided. We rely on two of the data tables: `Games.csv`, which provides game-level information including team identifiers and the *Last Stone First End* (LSFE) indicator that determines initial hammer possession, and `Ends.csv`, which records end-level outcomes such as points scored by each team and whether the Power Play was activated.

`Games.csv` contains game-level information for 344 mixed doubles matches, including unique game identifiers and the Last Stone First End (LSFE) indicator, which specifies which team begins the match with hammer. `Ends.csv` provides end-level scoring data, with one row per team per end, covering 2,637 total ends. Among these, 883 ends involve Power Play usage, while the remaining 1,754 ends are played under normal rules.

2.1 End-Level Outcome Definition

We treat the end as the fundamental unit of analysis. For each end in which a team has the hammer, we summarize the outcome by the

end-level score differential := points scored by the hammer team−points scored by the non-hammer team.

Zero or negative values indicate that the hammer team failed to score and therefore retained the hammer in the subsequent end.

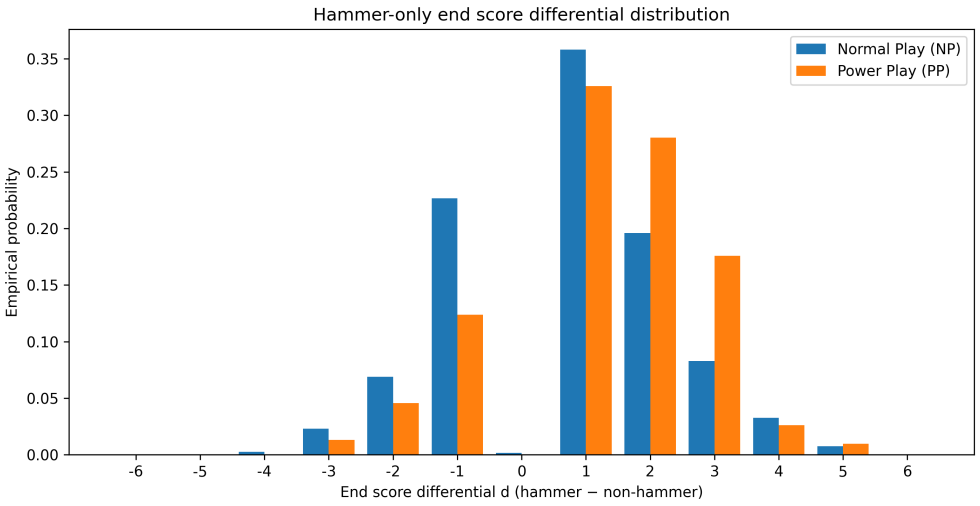


Figure 1: Empirical distribution of score differentials for the hammer team.

2.2 Normal Play versus Power Play Outcomes

End-level outcomes are grouped according to whether the Power Play was used, i.e. normal-play ends vs Power Play ends. From the .csv files, we can obtain the empirical distributions (See Figure 1) of end-level score differentials for each group, which reflect systematic differences in scoring behavior across the two settings. Power Play ends generally exhibit greater variability in outcomes. In the modeling framework introduced in the next section, these empirical distributions serve as the stochastic transition inputs governing how the game evolves under different strategic choices.

3 Modeling the Decision Problem

We formalize Power Play timing as a finite-horizon stochastic decision problem. The game evolves in discrete stages indexed by the number of ends remaining, and uncertainty arises from the stochastic scoring outcomes of each end. At the beginning of each end, the game state is represented by a tuple

$$(t, s, h, p_{\text{you}}, p_{\text{opp}}),$$

where t denotes the number of ends remaining, s is the current score differential (focal team minus opponent), h indicates hammer possession, and p_{you} and p_{opp} indicate whether the Power Play remains available to each team. Here, the subscript "you" refers to the focal team for which win probabilities are evaluated, while "opp" refers to the opposing team. The team with hammer chooses an action $a \in \{\text{NP}, \text{PP}\}$, subject to Power Play availability. Conditional on a, the end-level score differential d is drawn from an action-specific empirical distribution $p_a(d)$ shown in Figure 1. The realized outcome induces a deterministic transition to the next state via updates to the score, hammer possession, and Power Play availability.

We evaluate strategies using a value function $V(t, s, h, p_{\text{you}}, p_{\text{opp}})$ defined as the probability that the focal team wins the match from the given state under optimal play. At terminal states ($t = 0$), V is determined by the final score differential. For $t > 0$, V satisfies a Bellman recursion: when $h = 1$, the focal team selects a to maximize the expected continuation value, while when $h = 0$, the opponent selects a to minimize it.

Backward induction over t yields both the win probability associated with each state and the corresponding optimal Power Play policy. The full dynamic programming formulation, including explicit Bellman equations and transition rules, is given in Appendix A.

4 Accounting for Uncertainty

The empirical distributions Figure 1 are estimated from past games and therefore contains statistical uncertainty due to sampling variation, particularly for Power Play ends because there are fewer such ends. We adopt a Bayesian model to quantify the related uncertainty of our strategic recommendations.

For each action $a \in \{\text{NP}, \text{PP}\}$, let $p_a = \{p_a(d)\}_{d \in \mathcal{D}}$ denote the distribution of end-level score differentials. We place a Dirichlet prior on p_a and update it using the observed counts from the data, yielding a posterior distribution for each p_a.

Uncertainty is propagated through the decision model via Monte Carlo sampling. Specifically, for each posterior draw $\left(p_{\text{NP}}^{(m)}, p_{\text{PP}}^{(m)}\right)$, we solve the dynamic program to obtain a value function $V^{(m)}$ and an associated optimal policy. Repeating this procedure produces posterior distributions over win probabilities and over the optimal Power Play decision at each game state.

These outputs allow us to report, for each state, the posterior mean and variability of the win probability, as well as the posterior probability that activating the Power Play is optimal. This distinguishes game situations where Power Play timing recommendations are robust from those where they are sensitive to data uncertainty. Formal details of the Bayesian model and posterior sampling procedure are provided in Appendix B.

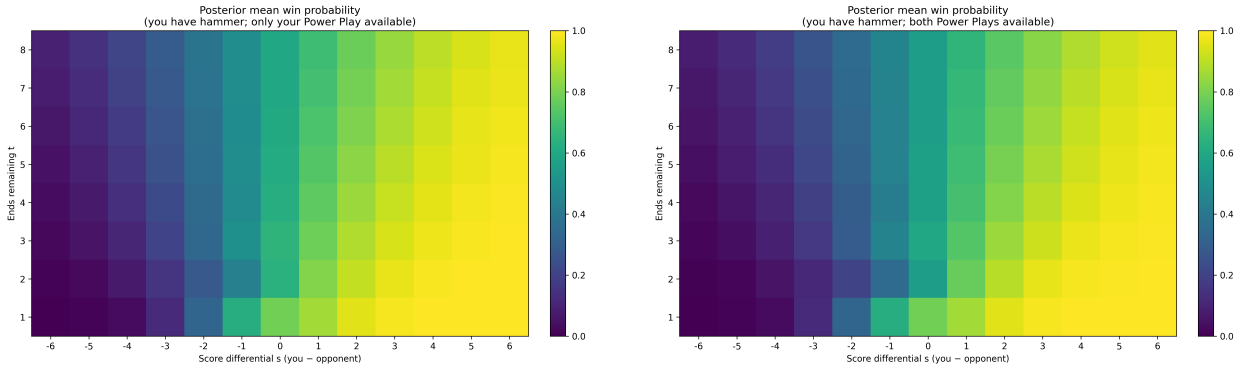
5 Results

5.1 Win Probabilities

Posterior mean win probabilities depend strongly on score differential, remaining ends, and hammer possession. For a fixed score state, win probabilities vary smoothly early in the game but change sharply late, reflecting the diminishing opportunity for future score swings. For example, when the score is tied ($s = 0$), win probabilities are near even with seven or eight ends remaining, but with one end remaining they shift rapidly depending on hammer possession. Similarly, when trailing by one point, the hammer team retains a meaningful chance of winning early in the game (e.g., $t = 6$ or $t = 7$), which declines quickly as the number of remaining ends decreases.

Hammer possession introduces a consistent asymmetry across the state space. For otherwise identical score and time states, the hammer team exhibits higher win probabilities, with the effect most pronounced in early and mid-game situations such as $t = 5$, $s = -1$ or $s = 0$. In late-game states, hammer possession becomes decisive: with one or two ends remaining, it largely determines whether a narrow deficit remains recoverable. These patterns motivate the subsequent analysis of Power Play decisions that explicitly leverage hammer advantage.

Figure 2 displays posterior mean win probabilities as functions of score differential and remaining ends, with darker purple colors indicating lower probabilities of winning and brighter yellow colors indicating higher probabilities. The horizontal axis represents the score differential, with negative values to the left and positive values to the right, while the vertical axis represents the number of ends remaining, with earlier game states at the top and later game states toward the bottom.



(a) Only the focal team's Power Play available.

(b) Both teams' Power Plays available.

Figure 2: Posterior mean win probability surfaces when the focal team has hammer, comparing Power Play availability regimes.

5.2 Optimal Power Play Timing

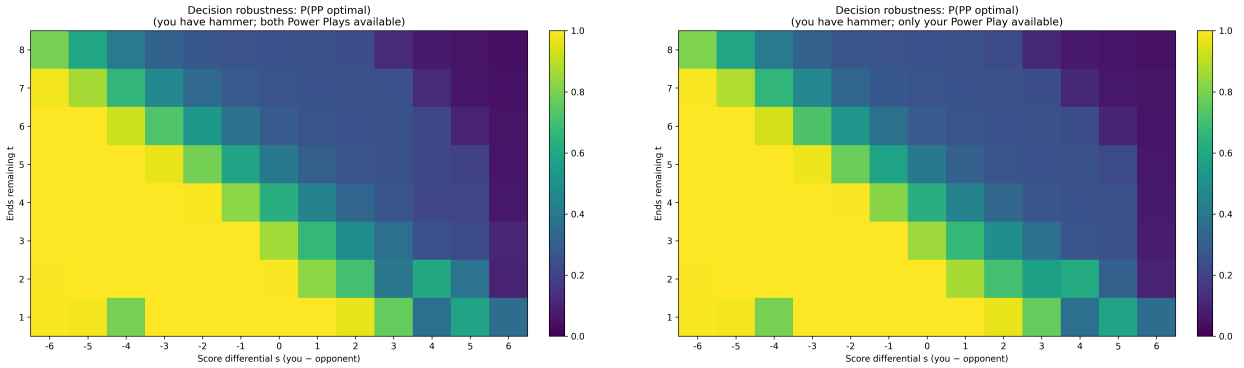
The optimal Power Play policy implied by the model displays well-defined regions of activation that depend jointly on score differential and remaining ends. In general, the model recommends activating the Power Play when the hammer team is trailing by small to moderate margins in early to mid-game situations. For instance, when trailing by two points with four ends remaining ($t = 4$, $s = -2$) or by three points with five ends remaining ($t = 5$, $s = -3$), the model recommends Power Play activation with near certainty. As the game progresses, the policy expands to include smaller deficits and even tied scores; for example, with one end remaining and a tied score ($t = 1$, $s = 0$), Power Play activation is still strongly favored. By contrast, when leading in late-game states such as $t = 2$, $s = 3$, the model typically recommends delaying or avoiding Power Play usage in order to reduce outcome variance.

Figures 3 and 4 display Power Play recommendations across all remaining ends and score differentials from -6 to 6, from the perspective of the hammer team. In Figure 3, colors represent the posterior probability that activating the Power Play is optimal, with dark purple indicating low probability and bright yellow indicating high probability; the surface varies continuously across states. Figure 4 presents the corresponding binary decision tables obtained by thresholding these probabilities at 0.5, where yellow indicates states in which the Power Play is recommended and purple indicates states in which normal play is preferred. Similar to earlier figures, the horizontal axis represents the score differential, with negative values to the left and positive values to the right, while the vertical axis represents the number of ends remaining, with earlier game states at the top and later game states toward the bottom.

5.3 Robustness and Decision Confidence

Posterior uncertainty provides important context for interpreting the model's recommendations. Figure 5 summarizes posterior standard deviations of win probabilities across score differentials and remaining ends when both Power Plays are available. Uncertainty is highest in balanced states and lowest when the score or remaining ends largely determine the outcome.

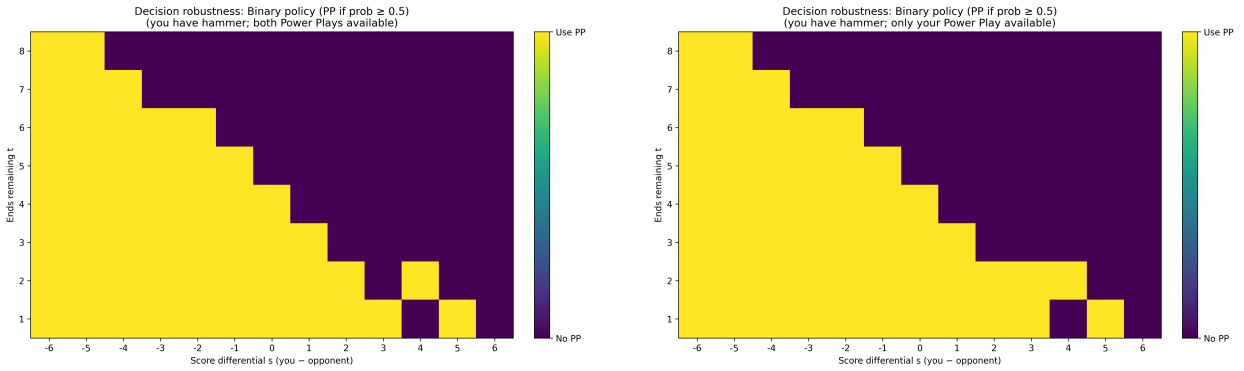
In early-game states with small score differentials, posterior standard deviations are moderate; for example, with seven ends remaining and a near-even score ($t = 7$, $s = \pm 1$), the standard deviation



(a) Both Power Plays available.

(b) Only the focal team's Power Play available.

Figure 3: Posterior probability that activating the Power Play is optimal, conditional on hammer possession, under different Power Play availability regimes.



(a) Both Power Plays available.

(b) Only the focal team's Power Play available.

Figure 4: Binary Power Play decision tables derived by activating the Power Play when the posterior probability of optimality exceeds 0.5.

is approximately 0.004–0.005. As the game progresses, uncertainty concentrates in close late-game states, peaking at approximately 0.016 when one end remains and the team is narrowly trailing (e.g., $t = 1$, $s = -2$).

By contrast, posterior uncertainty is negligible in lopsided states. When the score differential is large or the game is effectively decided, such as $t = 1$ with $|s| \geq 5$, posterior standard deviations approach zero.

The posterior probability that activating the Power Play is optimal further distinguishes robust from sensitive decisions. In several trailing states, including $t = 4$, $s = -2$ and $t = 5$, $s = -3$, this probability exceeds 0.9, indicating stable recommendations. In contrast, states such as $t = 5$, $s = 0$ or $t = 4$, $s = 1$ exhibit probabilities closer to one-half, reflecting genuine strategic ambiguity. Together, these results clarify when Power Play decisions can be made confidently and when flexibility remains appropriate.

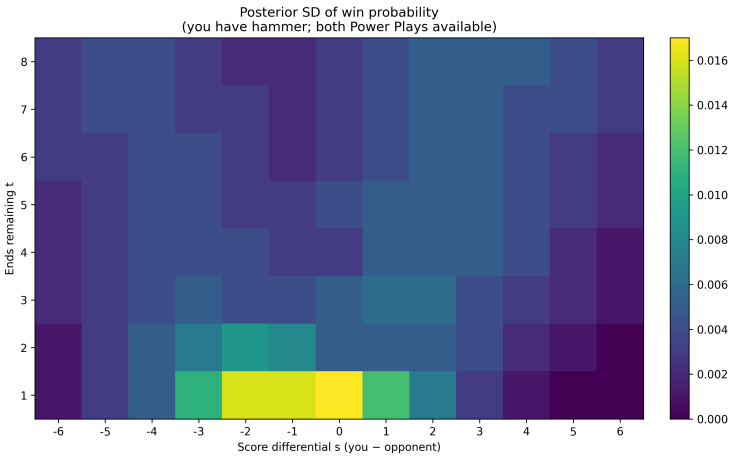


Figure 5: Standard deviation of the win probabilities when both power plays are available.

6 Limitations and Future Improvements

Our analysis relies on several simplifying assumptions. We treat end-level outcome distributions under normal play and the Power Play as stationary across teams, matches, and game stages, and observed Power Play usage may exhibit selection effects if teams preferentially activate it in favorable situations. Both factors can bias empirical outcome estimates.

Future extensions could relax these assumptions by incorporating team- or opponent-specific outcome distributions, time-varying dynamics, or richer data sources such as shot-level information or player-specific execution variability, enabling more personalized Power Play strategies within the same decision-theoretic framework.

7 Conclusion

This study develops a data-driven, decision-theoretic framework for optimal Power Play timing in mixed doubles curling. By modeling the match as a finite-horizon stochastic game and evaluating strategies through backward induction, we provide prescriptive guidance on when activating the Power Play maximizes the probability of winning.

A key contribution of the framework is its explicit treatment of uncertainty. By propagating posterior uncertainty in end-level outcome distributions through the decision model, we distinguish robust strategic recommendations from those that are sensitive to data limitations. This uncertainty-aware perspective is particularly important in settings such as mixed doubles curling, where Power Play usage is infrequent and empirical samples are limited.

Beyond the specific application to curling, the framework introduced here applies more broadly to sports and competitive settings involving limited-use strategic options and sequential decision-making under uncertainty. The results demonstrate how combining empirical data with formal decision models can yield actionable strategic insights while transparently accounting for uncertainty.

Appendix

A Dynamic Programming Formulation

We formalize Power Play timing as a finite-horizon stochastic game with state space

$$(t, s, h, p_{\text{you}}, p_{\text{opp}}),$$

where $t \in \{0, 1, \ldots, T\}$ denotes the number of ends remaining, $s \in \mathbb{Z}$ is the score differential (focal team minus opponent), $h \in \{0, 1\}$ indicates hammer possession ($h = 1$ if the focal team has hammer), and $p_{\text{you}}, p_{\text{opp}} \in \{0, 1\}$ indicate Power Play availability for the focal team and the opponent, respectively.

A.1 Actions and Transitions

When the team with hammer acts, it selects an action

$$a \in \mathcal{A}(p) = \begin{cases} \{\text{NP}, \text{PP}\}, & p = 1, \\ \{\text{NP}\}, & p = 0, \end{cases}$$

where p denotes the acting team's Power Play availability. Conditional on action a, the end-level score differential d is drawn from an empirical distribution $p_a(d)$ supported on a finite set $\mathcal{D}$.

State transitions are deterministic given d:

- If $h = 1$ (focal team has hammer),

$$s' = s + d, \quad h' = \mathbb{I}(d \leq 0), \quad p'_{\text{you}} = p_{\text{you}} - \mathbb{I}(a = \text{PP}), \quad p'_{\text{opp}} = p_{\text{opp}}.$$

- If $h = 0$ (opponent has hammer),

$$s' = s - d, \quad h' = \mathbb{I}(d > 0), \quad p'_{\text{you}} = p_{\text{you}}, \quad p'_{\text{opp}} = p_{\text{opp}} - \mathbb{I}(a = \text{PP}).$$

A.2 Value Function and Bellman Recursion

Let $V(t, s, h, p_{\text{you}}, p_{\text{opp}})$ denote the probability that the focal team wins the match from the given state under optimal play. The terminal condition is

$$V(0, s, h, p_{\text{you}}, p_{\text{opp}}) = \begin{cases} 1, & s > 0, \\ 0.5, & s = 0, \\ 0, & s < 0. \end{cases}$$

For $t \geq 1$, the Bellman recursion is given by

$$V(t, s, 1, p_{\text{you}}, p_{\text{opp}}) = \max_{a \in \mathcal{A}(p_{\text{you}})} \sum_{d \in \mathcal{D}} p_a(d)\, V(t - 1, s', h', p'_{\text{you}}, p'_{\text{opp}}),$$

when the focal team has hammer, and

$$V(t, s, 0, p_{\text{you}}, p_{\text{opp}}) = \min_{a \in \mathcal{A}(p_{\text{opp}})} \sum_{d \in \mathcal{D}} p_a(d)\, V(t - 1, s', h', p'_{\text{you}}, p'_{\text{opp}}),$$

when the opponent has hammer, where $(s', h', p'_{\text{you}}, p'_{\text{opp}})$ are defined by the transition rules above.

Backward induction over t yields both the value function and the optimal Power Play policy.

A.3 Solving the Bellman Recursion

The Bellman recursion is solved by backward induction over the number of remaining ends. Because the horizon is finite and the state space is discrete, this procedure is exact and computationally tractable.

We proceed as follows.

Step 1: Discretize the state space. We enumerate all feasible states $(t, s, h, p_{\text{you}}, p_{\text{opp}})$, where t ranges from 0 to the maximum number of ends, score differentials s are restricted to a finite window observed in the data, and h, p_{you}, and p_{opp} are binary. States outside this window are truncated to boundary values, which has negligible effect on results due to their extremely low probability.

Step 2: Initialize terminal values. At terminal states ($t = 0$), the value function is initialized according to the final score differential:

$$V(0, s, h, p_{\text{you}}, p_{\text{opp}}) = \begin{cases} 1, & s > 0, \\ 0.5, & s = 0, \\ 0, & s < 0. \end{cases}$$

These values serve as the boundary condition for the recursion.

Step 3: Backward induction. For $t = 1, 2, \ldots, T$, the value function is computed recursively using values from time $t - 1$. For each state $(t, s, h, p_{\text{you}}, p_{\text{opp}})$:

- If $h = 1$ (focal team has hammer), we compute the expected continuation value for each feasible action $a \in \mathcal{A}(p_{\text{you}})$ by averaging the value function at successor states with respect to the outcome distribution $p_a(d)$. The action attaining the maximum expected value is selected.
- If $h = 0$ (opponent has hammer), we compute the same expectations over $a \in \mathcal{A}(p_{\text{opp}})$ and select the action attaining the minimum expected value, reflecting adversarial play.

This step defines both the value function $V(t, s, h, p_{\text{you}}, p_{\text{opp}})$ and the associated optimal action at each state.

Step 4: Policy extraction. The optimal Power Play policy is obtained by recording, for each state in which the acting team has hammer and the Power Play is available, whether the maximizing action is normal play or Power Play. This produces a deterministic policy mapping game states to actions for each posterior draw of the outcome distributions.

Step 5: Integration with posterior sampling. When uncertainty is incorporated (Appendix B), Steps 2–4 are repeated independently for each posterior draw of the end-level outcome distributions. The resulting collection of value functions and policies is then aggregated to obtain posterior summaries, such as the probability that activating the Power Play is optimal at a given state.

This backward induction procedure guarantees that all strategic decisions account for their full downstream consequences, yielding optimal policies for both teams under the assumed model.

B Bayesian Modeling and Posterior Propagation

For each action $a \in \{\text{NP}, \text{PP}\}$, let $\mathcal{D}$ denote the finite set of possible end-level score differentials and let

$$p_a = \big(p_a(d)\big)_{d \in \mathcal{D}}$$

denote the corresponding outcome distribution. We model observed end-level outcomes using a multinomial sampling model with unknown parameter p_a.

We place a Dirichlet prior on p_a,

$$p_a \sim \text{Dirichlet}(\alpha_a),$$

where α_a is a vector of positive concentration parameters. Let $n_a(d)$ denote the number of observed outcomes equal to d under action a, with $n_a = \big(n_a(d)\big)_{d\in\mathcal{D}}$. By Dirichlet–multinomial conjugacy, the posterior distribution is

$$p_a \mid \{n_a(d)\}_{d\in\mathcal{D}} \sim \text{Dirichlet}(\alpha_a + n_a),$$

with posterior mean

$$\mathbb{E}[p_a(d) \mid \text{data}] = \frac{\alpha_a(d) + n_a(d)}{\sum_{d'\in\mathcal{D}}(\alpha_a(d') + n_a(d'))}.$$

Uncertainty is propagated through the decision model via Monte Carlo sampling. For $m = 1, \ldots, M$, we draw independent samples

$$p_a^{(m)} \sim \text{Dirichlet}(\alpha_a + n_a), \quad a \in \{\text{NP}, \text{PP}\},$$

each of which defines a distinct set of transition probabilities for end-level scoring outcomes.

For each posterior draw $(p_{\text{NP}}^{(m)}, p_{\text{PP}}^{(m)})$, we solve the dynamic programming recursion described in Appendix A to obtain a value function $V^{(m)}$ and an associated optimal Power Play policy $\pi^{(m)}$. Aggregating over posterior draws yields state-wise posterior summaries, including the posterior mean and variance of the win probability and the posterior probability that activating the Power Play is optimal at a given game state.

This approach provides a principled mechanism for incorporating finite-sample uncertainty in empirical outcome distributions into both win probability estimation and strategic decision-making.

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